TEST 101

Fall Semester, AY 2024-2025

Midterm Exam

November 1, 2024

Instructions:

1. You have 60 minutes to complete the test
2. For numerical questions, select the option that is the closest to your computation
3. Please read all questions carefully and make sure you understand the facts before you begin answering
4. For multiple choice and true & false questions, choose only one answer per question. Questions having more than one choice will be discarded during grading
5. The university's policy regarding academic dishonesty will be strictly enforced. You are bound by King Saud University's Honor Code
6. Good luck!

I have read and understood all the instructions

Name: _____

University ID: _____

Part	TF	MC	Total
Grade	5	3	8
Score			

I) [5 points] *The following section contains statements, evaluate each statement whether it is true or false. Each evaluation attempt is worth 1 point.*

1. Example for a MC question
   ◯ True ◯ False
2. Example for a MC question
   ◯ True ◯ False
3. Example for a TF question
   ◯ True ◯ False
4. Example for a MC question
   ◯ True ◯ False
5. Example for a TF question
   ◯ True ◯ False

II) [3 points] *The following section contains multiple choice questions. Please choose the correct (or the closest) choice per question. Each question is worth 1 point.*

1. You manage a portfolio that has the following information:

Stock	Weight	Alpha	Beta
Taiba	10%	1.25%	0.8
Rajhi	90%	0.5%	1.6

The current t-bill rate is 2%, and the market risk premium is 4%. Using the index model, the expected return on this portfolio is:

A. 12.24%
B. 6.69%
C. 3.79%
D. 9.02%

2. Most of the initial public offerings (IPOs) of equity securities are bought by _____.

A. individual investors
B. institutional investors
C. day traders
D. the firm's current shareholders

3. Most of the initial public offerings (IPOs) of equity securities are bought by _____.

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Table 1: Formulas

$$\text{Discount from Face Value} = r_{BD} \times \frac{n}{360}$$

$$\text{Price} = \text{Face Value} \times \left[1 - \left(r_{BD} \times \frac{n}{360}\right)\right]$$

$$r_{BD} = \frac{\text{Face Value} - P}{\text{Face Value}} \times \frac{360}{n}$$

$$r_{BEY} = \frac{\text{Face Value} - P}{P} \times \frac{365}{n}$$

$$r_{EAY} = \left[1 + \frac{\text{Face Value} - P}{P}\right]^{\frac{365}{n}} - 1$$

$$\text{Price Weighted Index}_t = \frac{\sum_{n=1}^{N} \text{Price}_{n,t}}{\text{Divisor}_t}$$

$$\text{Market Value}_{n,t} = \text{Stock Price}_{n,t} \times \text{No. Shares}_{n,t}$$

$$\text{Equal Weighted Index Return} = \frac{1}{n} \times \sum_{n=1}^{N} R_{n,t}$$

$$\text{Market Value Weighted Index Return}_t = \frac{\sum_{n=1}^{N} \text{Market Value}_{n,t}}{\sum_{n=1}^{N} \text{Market Value}_{n,t-1}} - 1 = \sum_{n=1}^{N} W_{n,t-1} \times R_{n,t}$$

$$W_{n,t-1} = \frac{\text{Market Value}_{n,t-1}}{\sum_{n=1}^{N} \text{Market Value}_{n,t-1}}$$

$$\text{Underpricing} = \frac{\text{First day closing Price} - \text{Offering Price}}{\text{Offering Price}}$$

$$\text{Bid-ask Spread} = \text{Price}_{Ask} - \text{Price}_{Bid}$$

$$\text{Margin} = \frac{\text{Investor's equity in the account}}{\text{Current value of Stock}}$$

$$\text{Max \% Investor can borrow} = (1 - IMR)$$

$$\frac{\text{Investor's equity in the account}}{\text{Current value of Stock}} \leq MMR$$

$$\text{Short Sale Margin} = \frac{\text{Investor's equity in the account}}{\text{Value of shares owed}}$$

$$NAV = \frac{\text{Market Value of Assets} - \text{Liabilities}}{\text{No. Shares Outstanding}}$$

$$\text{Premium (Discount)} = \frac{\text{Price} - NAV}{NAV}$$

$$\text{Offering Price} = \frac{NAV}{1 - \text{load}}$$

$$\text{Rate of Return}_t = \frac{NAV_t - NAV_{t-1} + \text{Income}_t + \text{Cap. Gains}_t}{NAV_{t-1}}$$

$$\text{Turnover} = \frac{\text{Assets that are replaced or sold}}{\text{Total Assets being Managed}}$$